

Shock (Medical)

Signs of Shock

- Shock is defined as inadequate perfusion of blood and oxygen to the brain, heart and other vital organs
- Hypotension (SBP $\leq$ 90)
- Tachycardia (except in neurogenic shock)
- Tachypnea / difficulty breathing / rales
- Syncope / near – syncope / Lightheadedness / dizziness
- Altered mental status / confusion / agitation
- Poor peripheral perfusion – cool extremities, decreased capillary refill, pale or gray skin color
 - Skin may be warm in some patients with septic shock

Types of Shock

- **Cardiogenic** – usually caused by acute myocardial infarction (AMI), including right ventricular (RV) infarct
 - Can also be caused by PE, medications (beta blockers, calcium channel blockers, etc.), cardiomyopathy (infection to heart muscle), cardiac tamponade
 - May cause pulmonary edema/JVD
- **Septic** – by infection, see Sepsis Alert Guideline
 - May be associated with fever or hypothermia
- **Neurogenic** – caused by injury or infection to spinal cord
 - Patients are usually hypotensive and bradycardic
- **Anaphylactic** – due to severe systemic allergic reaction
- **Hypovolemic** – usually hemorrhagic
 - Usually due to trauma, GI bleed, AAA, ectopic pregnancy
 - Low BP is a late finding in hypovolemic shock

EMT/AEMT

- Follow General Medical Care Guideline
- Follow Adult Resuscitation Guideline

EMT/AEMT - Expanded Scope

- **NS/LR** 500ml IO bolus if unable to tolerate fluid by mouth and meeting criteria for shock

Paramedic

If SBP $\leq$ 90 or other signs/symptoms of shock, give fluids and consider pressor to maintain SBP $>$ 90 or improve signs/symptoms of shock

- **NS/LR** start fluid bolus IO/IV, amount of fluid to be given depends on clinical condition and patient response to fluids
 - Hypovolemic shock generally should have fluid boluses only, pressors may be considered
 - Septic shock should have fluids and pressors started simultaneously
 - Cardiogenic shock should have fluid bolus challenge (250mL) prior to pressor administration, then fluids and pressors simultaneously

AND (if indicated)

- **Epinephrine Push Dose** (1:100,000) 10-20 mcg (1-2 mL) IO/IV every 2-3 minutes PRN (max total 100 mcg / 10 mL)
 - Consult Medical Director for further doses

OR

- **Epinephrine Drip** (0.05 to 5 mcg/kg/min) titrate to effect, (mix Epi [1:1,000] 2mg in 500 mL NS bag)

